

Reconstructing the Clebsch code and its golden orientation from its deep-hole syndrome locus

Tavis Rudd

July 2026

*From deep holes, a cubic **takes shape**; its companion finds its bearings,
the carrier stands fixed while their shadows move; from minimum words, the plane returns,
and the scattered shadows gather home.*

Abstract

Let A be a six-arc in $\text{PG}(2, 11)$, and let $\mathcal{U}(A)$ be the projective points lying on no chord of A . For the associated $[6, 3, 4]_{11}$ MDS code, $\mathcal{U}(A)$ is its projective deep-hole syndrome locus. We prove that $\mathcal{U}(A)$ lies on a conic if and only if A is projectively equivalent to the Clebsch hexagon; in that case $\mathcal{U}(A)$ is exactly a nonsingular conic. Thus nearest-codeword data reconstruct the non-GRS Clebsch code up to monomial equivalence: the projective deep-hole locus is a recognition invariant whose metric boundary data recover the parity-check geometry. The reconstructed conic then determines its polarity, and Dye’s theorem identifies the stabilizer as A_5 ; none of this geometry is assumed.

Nearest-codeword ambiguity then reconstructs an unordered orientation torsor on six axes. Its signed orbital operator satisfies $B^2 = 5I$, and triangle holonomy gives the support cubic through $c_{ijk} = B_{ij}B_{jk}B_{ki}$; equivalently, this cubic is the sole nonsymmetric term in the diagonal determinant pencil of B . Thus the same syndrome and support data recover both the code and its golden orientation; coset-leader ambiguity recovers not only incidence and symmetry but the integral quadratic order $\mathbf{Z}[B] \simeq \mathbf{Z}[\sqrt{5}]$.

The proof uses a universal chord-defect identity and a partial-cover bound for rigidity, then decoder ambiguity and the orbital pentagon for orientation. As a secondary uniform consequence, any k -arc whose uncovered locus is a nonsingular conic has q odd, with $2k - 3 \leq q \leq (k(k - 1) + 3)/3$. Thus for each fixed k , the all-field existence problem reduces to finitely many field orders.

Keywords. Clebsch hexagon; MDS code; deep hole; arc; conic; monomial equivalence; two-graph; golden ratio.

2020 Mathematics Subject Classification. 51E21, 94B05, 51E22, 05B25.

1 Introduction

The complete nearest-codeword structure of a code can retain geometry that is absent from its definition. The example here is a non-GRS $[6, 3, 4]_{11}$ MDS code. Its maximum-distance syndrome directions form a conic, and that coarse locus already determines the projective code: its six parity-check columns are projectively equivalent to the Clebsch hexagon, with the associated polarity and A_5 symmetry. No conic, polarity, group action, or classical hexagon structure is assumed. Thus the problem is an inverse one: when does the projective deep-hole locus determine an MDS code, and how much of its geometry can nearest-codeword data recover?

Series perspective. *Clebsch: Rigidity from Sparse Shadows* asks when incomplete invariants determine the structures that produced them. The present paper begins with deep-hole syndrome data and reconstructs the code, conic geometry, and golden orientation. Papers I and III reconstruct marked conference companions, whereas Paper II reconstructs a signed chordal companion. Paper V proves that they share one invariant pencil and six-axis carrier and, after marking a chordal line, recover each other with orientation [22]. Paper IV is the parallel lower branch: minimum-word pairs reconstruct a marked conic plane and polarity, with no cubic identification asserted. Figure 1 records this division.

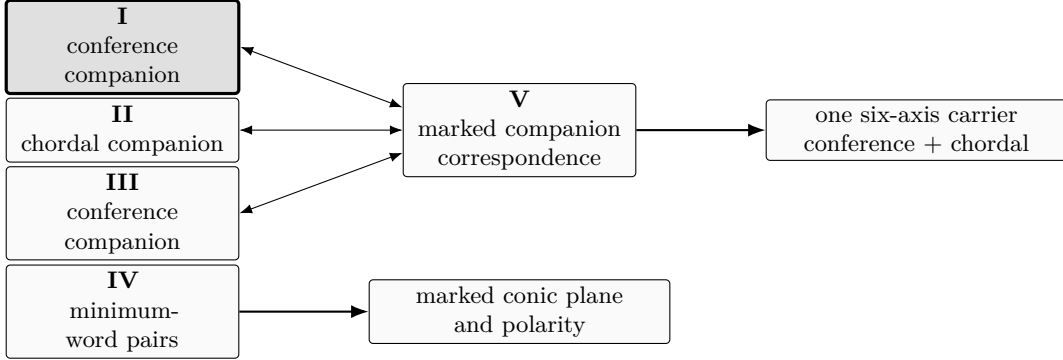


Figure 1: Series map. Upper double arrows are the marked transports and returns proved in Paper V; its right arrow is the common-carrier theorem. The lower solid arrow is Paper IV’s independent reconstruction.

Let C be a linear $[n, k, d]_q$ code with covering radius ρ and full-rank parity-check matrix H . For a received word v , write $\sigma(v) = Hv^\top$ and

$$w(s) = \min\{\text{wt}(e) : He^\top = s\}.$$

Then $d(v, C) = w(\sigma(v))$, and v is a *deep hole* precisely when this minimum is ρ . The syndrome s specifies one coset of C , whose coset leaders are its weight- $w(s)$ representatives. Because $w(\lambda s) = w(s)$ for $\lambda \neq 0$, maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\text{proj}}(C) = \{[s] \in \text{PG}(n - k - 1, q) : w(s) = \rho\}.$$

Thus one projective direction represents $q - 1$ distinct nonzero syndrome cosets, while each coset contains $|C|$ received words.

For an MDS code of redundancy $r = n - k$, the parity-check columns form a projective arc in $\text{PG}(r - 1, q)$, and $w(s)$ is the least number of columns whose span contains $[s]$. Adjoining a new column preserves the MDS property exactly when its direction avoids every hyperplane spanned by $r - 1$ existing columns. In the redundancy-three plane case used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. We write $\mathcal{U}(A)$ for this extension locus. If $\mathcal{U}(A) \neq \emptyset$, then the associated code has covering radius three and $\mathcal{U}(A) = \mathcal{D}_{\text{proj}}(C)$. This dictionary is classical in the theory of complete arcs and is developed for MDS codes by Davydov, Marcugini, and Pambianco [8]; their Theorem 6.3 supplies the leader count used in Section 6. See also Hirschfeld [17] for the plane arc/covering-radius setting. We call A *conic-filling* when $\mathcal{U}(A)$ is the full $\mathbb{F}_q$ -rational point set of a nonsingular conic.

Here and in Theorem 1.1, a *conic* is the projective zero set of a nonzero homogeneous quadratic form; it may be degenerate unless stated otherwise.

Theorem 1.1 (Rigidity theorem). *For a six-arc $A \subset \text{PG}(2, 11)$, the following are equivalent:*

- (i) *the uncovered locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);*
- (ii) *$\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$ -points of a nonsingular conic;*
- (iii) *A is $\text{PGL}(3, 11)$ -equivalent to the Clebsch hexagon.*

More precisely, for a fixed nonsingular conic $\mathcal{C}$ the six-arcs satisfying $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_{11})$ form one orbit under $\text{Stab}_{\text{PGL}(3, 11)}(\mathcal{C})$ (Corollary 4.2). That is the exact geometric sense in which the syndrome locus reconstructs the six-column realization.

The proof reconstructs the hexagon from conic containment alone. A line bound, a chord-defect identity, and Dye’s Brianchon theorem give the conceptual implication. The same geometry then gives an exact distance oracle and all nearest-codeword multiplicities; the resulting ambiguity strata reconstruct the Brianchon points, the self-polar triangles, and the invariant support bipartition. On the reconstructed twelve-point syndrome locus, the two five-valent orbitals recover a signed operator on the six antipodal axes. Triangle holonomy identifies its switching class with the support bipartition and recovers the integral order $\mathbf{Z}[\sqrt{5}]$.

This orientation is the second structural pillar. With the normalization of Theorem 8.1, its signed operator and support cubic satisfy

$$B^2 = 5I, \quad c_{ijk} = B_{ij}B_{jk}B_{ki},$$

$$\det(B + \text{diag}(x_0, \dots, x_5)) = e_6(x) - e_4(x) + 5e_2(x) - 125 - 2C(x).$$

Thus triangle holonomy and the diagonal determinant pencil recover the same orientation torsor directly from nearest-codeword data.

A secondary general theorem gives a uniform field window for conic-filling arcs. For an arbitrary conic-filling k -arc, the chord moments and a partial-cover theorem force q odd and $2k - 3 \leq q \leq (k(k - 1) + 3)/3 < \binom{k}{2}$. For a Clebsch hexagon K , the same chord-defect identity gives $|\mathcal{U}(K)| = q^2 - 14q + 45$, with off-conic excess $(q - 4)(q - 11)$ when $q \equiv 3 \pmod{4}$.

The configuration goes back to Clebsch [7, p. 336]. Edge constructed its finite-plane form and determined its order-60 stabilizer [12, §§29–32]; Dye proved projective uniqueness and determined the stabilizer over a general field [11, Theorems 1 and 3, pp. 275–278]; Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [28]. Deep holes of Reed–Solomon codes have both an algorithmic history [14, 6] and a geometric classification in the projective case [30]. Kaipa makes the deep-hole/MDS-extension equivalence explicit for Reed–Solomon codes [18], while Wu, Ding, and Chen formulate a corresponding extension criterion for general MDS codes and dual deep holes [29]. Those results begin with a fixed code and characterize its extensions. The inverse statement here instead reconstructs a non-GRS projective code from its entire maximum-distance syndrome locus.

What is new and what is not. The hexagon, its uniqueness and stabilizer, and its complete-exterior-set interpretation are classical [7, 12, 11, 4, 28]. We prove the conic-locus identification of Proposition 6.1 directly here, and Proposition 2.2 reproves the classical existence, orbit, polarity, and stabilizer statements from one normal form over $\mathbf{Z}[\varphi]$. The new contribution is the symmetry-free arbitrary-quadratic-containment coding criterion, the decoder reconstruction and support bipartition, the uniform conic-filling window, and the concurrence spectrum that refines Dye’s bound. The reconstructed support bipartition and

syndrome cover also determine a balanced signed two-graph and its golden operator; their triangle products recover a cubic determinant line. The underlying six-nodal S_5 -symmetric cubic is known [5]; what is proved here is its intrinsic recovery from the code's syndrome and support data, together with the integral continuation order. A separate computational companion [21] proves sharper finite consequences: a degree-three characterization and numerical gap at $q = 11$, uniqueness of $q = 11$ among conic-filling six-arcs, classification through eight points, and the original computational form of an exact $q = 13$ binary incidence-code theorem. Paper IV gives the structural and reproducible account of that $q = 13$ consequence [26].

Paper II studies what the conic matching quotient remembers [23]; Paper III supplies the characteristic-zero golden source [25]; and Paper IV gives the independent minimum-word reconstruction [26]. The reconstructed operator also has uses beyond this inverse problem: the unnumbered Golden quantum-statistics companion studies its exchange-statistical interpretation [24].

2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C} : XZ = Y^2$$

in $\text{PG}(2, 11)$, with its 12 points identified with $\text{PG}(1, 11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\text{PGL}(3, 11)$ is $\text{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

A convenient representative is

$$A = \{(1, 10, 0), (1, 9, 1), (1, 4, 7), (1, 8, 5), (0, 1, 4), (1, 1, 7)\}.$$

It is a six-arc disjoint from $\mathcal{C}$; its fifteen joins also avoid $\mathcal{C}$. The invariant construction in the following definition explains its symmetry.

Definition 2.1 (Clebsch hexagon). Following Dye, a *Clebsch hexagon* over a field is a six-arc with exactly ten nonvertex triple-chord concurrence points, called its *Brianchon points*.

At $q = 11$ there is also a group-theoretic construction. Let $A_5 \subset \text{PGL}_2(11)$ be an icosahedral subgroup; its six Sylow 5-subgroups each fix two points of $\mathcal{C}$, and the poles of the six chords joining these fixed pairs form a six-arc with ten Brianchon points [12, §§29–32]. The two descriptions agree over $\mathbb{F}_{11}$: by Proposition 2.2 the six-arcs with ten Brianchon points form a single $\text{PGL}_3(11)$ -orbit, which therefore contains the constructed arc.

For a nonsingular conic over an odd field, an *external point* lies on two tangents and an *internal point* on none; an *exterior*, or *passant*, line is disjoint from the conic. At $q = 11$, Edge's six vertices are external, their fifteen joins are exterior, and those joins concur three at a time at ten internal Brianchon points [12, §§29–32]. This reading is specific to order eleven, and we use it only there. At every odd order the six vertices share one point type and the ten Brianchon points share one point type, but which type occurs depends on the order: at $q = 19$ the two types are interchanged, the vertices being internal and the Brianchon points external. Tagged to $q = 11$, Edge's reading makes the hexagon a complete exterior set in the terminology of Blokhuis, Seress, and Wilbrink [4, pp. 143, 146]. Up to their stated notion of isomorphism, their $q = 11$ census records one six-arc type and also a Pasch configuration, but its four collinear triples prevent it from defining an MDS code. This supplies the classical complete-exterior-set predecessor, without our identifying their

isomorphism quotient with a particular projective stabilizer action. Our rigidity argument starts instead from the weaker hypothesis that $\mathcal{U}(A)$ lies on an arbitrary quadratic; it proves that the quadratic is nonsingular, that containment is equality, and that the arc lies in one projective orbit. Thus exteriority alone gives only $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; the arc hypothesis selects the Clebsch branch, and Proposition 6.1 proves equality.

A second close predecessor is the elliptic-line-cover result of Blokhuis, Brouwer, and Szőnyi. Immediately after their Proposition 1.6 they report, from a computer search, that at $q = 11$ the extremal cover of the complement of a fixed conic by fifteen elliptic lines is unique up to isomorphism [3, following Proposition 1.6]. Whenever $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_{11})$, the fifteen chords of A form precisely such a cover. Their published statement starts with the fixed conic and the line cover; it does not formulate the implication from containment in an arbitrary quadratic, prove that the containing quadratic is nonsingular and filled, or identify six arc vertices at the projective/monomial-equivalence level. We use their lower bound for the uniform field window, but the order-eleven rigidity proof does not use their computational uniqueness.

The displayed hexagon is the arc K_2 of Storme and Van Maldeghem [28, Props. 11–12] and Dye’s synthetic hexagon when 5 is a square in the base field [11, Theorem 1, pp. 275–276]. The following normal form carries the classical package uniformly in the field and makes every step an identity in $\mathbf{Z}[\varphi]$.

Proposition 2.2 (Golden normal form and uniform rigidity). *Let K be a field of odd characteristic and let φ satisfy $\varphi^2 = \varphi + 1$. A six-arc in $\text{PG}(2, K)$ has ten Brianchon points if and only if 5 is a square in K and the arc is $\text{PGL}_3(K)$ -equivalent to*

$$X_\varphi = \{(1:0:0), (\varphi:1:1), (0:1:0), (1:\varphi:1), (0:0:1), (1:1:2-\varphi)\}.$$

Such configurations therefore form one $\text{PGL}_3(K)$ -orbit. Their five self-polar triangles admit exactly one polarity, represented by

$$S = \begin{pmatrix} 2\varphi - 1 & -1 & -\varphi \\ -1 & 2\varphi - 1 & -\varphi \\ -\varphi & -\varphi & 3\varphi + 1 \end{pmatrix}, \quad \det S = 4\varphi,$$

and already determined by any two of those triangles. If $\text{char } K \neq 5$, the projective stabilizer is A_5 , acting on the six vertices as $\text{PSL}_2(5)$ acts on the projective line over $\mathbb{F}_5$; in characteristic five it is S_5 , acting as $\text{PGL}_2(5)$. Finally, the associated quadratic form takes values of norm -5 on the six vertices and of norm -9 on the ten Brianchon points: the vertices lie off the associated conic exactly when $\text{char } K \neq 5$, and the Brianchon points lie off it exactly when $\text{char } K \neq 3$.

Proof. Chords through an off-vertex concurrence are pairwise disjoint, so a concurrence is carried by one of the fifteen perfect matchings of the six vertices. At the maximum of ten, Dye’s equality analysis leaves five non-concurrent matchings forming a one-factorization [11, §2.2, pp. 274–275]. Label so that this is a fixed one-factorization and normalize $P_1 = (1:0:0)$, $P_3 = (0:1:0)$, $P_5 = (0:0:1)$, with the unit point as the first concurrence and $P_2 = (x:1:1)$, $P_4 = (1:y:1)$, $P_6 = (1:1:z)$. The ten concurrence determinants then reduce to $x = y$, $x + z = 2$, and $xyz = 1$, hence to $x^2(2 - x) = 1$, that is,

$$(x - 1)(x^2 - x - 1) = 0.$$

The root $x = 1$ places P_2 on a chord of the remaining vertices and is excluded by the arc condition, so $x = y = \varphi$ and $z = 2 - \varphi$; the quadratic has a root exactly when its discriminant

5 is a square. Conversely, all twenty triple determinants of X_φ reduce in $\mathbf{Z}[x]/(x^2 - x - 1)$ to ± 1 , $\pm(x - 1)$, $\pm(x - 2)$, or x , none of which vanishes in odd characteristic, and the five non-concurrence determinants reduce to $\pm 2(x - 1)$; thus X_φ is a six-arc with concurrence count ten. When 5 is a nonzero square the two roots φ and $\bar{\varphi} = 1 - \varphi$ give the same projective class: the relabelling (3 4 6 5) preserves the concurrence pattern and is realized by

$$G = \begin{pmatrix} -1 & 1 & 0 \\ 0 & \bar{\varphi} & \varphi \\ 0 & 1 & 0 \end{pmatrix}, \quad G \cdot X_\varphi = X_{\bar{\varphi}}, \quad \det G = \varphi - \bar{\varphi}.$$

For the stabilizer, the subgroup of S_6 preserving the five non-concurrent matchings has order 120, acting on them as S_5 through the outer automorphism of S_6 . When the roots are distinct, recording which root the normalization produces maps this subgroup onto the two roots, with fibres the cosets of the induced projective stabilizer; the displayed G shows the map is onto, so that stabilizer has index two and is A_5 . In characteristic five the two roots coalesce. The same normalization then realizes all 120 concurrence-pattern symmetries as projectivities of X_φ . Concretely, with $\varphi = 3 \in \mathbb{F}_5$, the matrix

$$R = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 3 & 0 & 2 \end{pmatrix}$$

permutes the six displayed vertices as (1 5 2 4) and induces a four-cycle on the five non-concurrent matchings. It is therefore in the odd coset of the pattern S_5 rather than its A_5 subgroup. Thus the stabilizer is the full S_5 .

For the polarity, self-polarity of the five triangles imposes fifteen linear conditions on the six-dimensional space of symmetric matrices. The displayed S satisfies all fifteen identically in $\mathbf{Z}[x]/(x^2 - x - 1)$, and the coefficient matrix has a five-by-five minor equal to $8(13 - 8\varphi)$, whose second factor has norm 1 and is therefore a unit of $\mathbf{Z}[\varphi]$. The rank is exactly five, so the solution space is the line through S ; that minor uses only rows from the first two triangles, so two triangles already determine S . Since $\det S = 4\varphi$ and $\varphi = 0$ would force $0 = 1$, the polarity is nondegenerate in every odd characteristic. Evaluating $v \mapsto v^T S v$ gives vertex values $2\varphi - 1$ (four times), $3\varphi + 1$, and $3\varphi - 4$, each of norm -5 , and Brianchon values $3\varphi - 3$, $6\varphi - 9$, and 3φ , each of norm -9 . A value vanishes exactly when its norm does, which gives the two characteristic conditions. $\square$

Remark 2.3 (Attribution). Proposition 2.2 reproves classical statements. Dye introduces the golden parameter j with $j^2 = j + 1$ as a root of the discriminant-five quadratic and fixes a canonical hexagon with vertices $(1, \pm j, 0)$, $(0, 1, \pm j)$, $(\pm j, 0, 1)$; his Theorem 1 gives existence exactly when the characteristic is not two and 5 is a square, together with transitivity of $\mathrm{PGL}_3(K)$ [11, §2.2, Theorem 1, pp. 274–276]. The unique polarity making the five triangles self-polar is his Theorem 2(iii) [11, pp. 277–278]. His Theorem 3 gives stabilizer A_5 except in characteristic five, where it gives S_5 [11, p. 278]; that two triangles in general position have a unique common self-polar conic is classical. What differs is the presentation: the triple-perspective normalization forces the factorization $(x - 1)(x^2 - x - 1) = 0$ rather than Dye's $(\lambda\mu\nu)^2 - 4\lambda\mu\nu - 1 = 0$, and every verification above is an identity over $\mathbf{Z}[\varphi]$ rather than a case analysis over K .

At $q = 11$ the conic of S has twelve points and is $\mathcal{C}$ in the displayed coordinates. That identification is order-specific: the concurrence count gives $|\mathcal{U}(A)| = 22 - 10 = 12 = q + 1$ only at $q = 11$, while at $q = 19$ the uncovered locus already has 140 points against a conic of 20. Dye's calculation also shows that the six vertices lie on no conic in characteristic 11

[11, Theorem 2, pp. 277–278; p. 281]. Storme and Van Maldeghem’s Propositions 11–12 identify this as the two-transitive A_5 -arc K_2 ; their Proposition 13 classifies the complete two-transitive arcs and omits this order-eleven six-arc [28, Props. 11–13]. It is therefore incomplete, so the associated redundancy-three code has covering radius three.

3 A universal chord-defect identity

Theorem 3.1 (Universal chord defect). *Let A be a k -arc in $\text{PG}(2, q)$ with $k \geq 4$. Let n_i count the off-arc points incident with exactly i chords, put*

$$r = \lfloor k/2 \rfloor, \quad t(A) = \sum_{i=3}^r \binom{i-1}{2} n_i,$$

and write $m = \binom{k}{2}$. Then

$$|\mathcal{U}(A)| = q^2 - (m-1)q + \left(m + 3\binom{k}{4} - k + 1 \right) - t(A),$$

$$0 \leq t(A) \leq 3\binom{k}{4} \frac{r-2}{r}, \quad k \leq 5 \implies t(A) = 0.$$

Proof. At a point off A , concurrent chords have pairwise disjoint endpoint pairs, so at most r chords occur. Counting chord–point incidences and pairs of disjoint chords gives

$$\sum_i i n_i = m(q-1), \quad \sum_i \binom{i}{2} n_i = 3\binom{k}{4}.$$

Since $\binom{i}{2} = (i-1) + \binom{i-1}{2}$, the number of covered off-arc points is $m(q-1) - 3\binom{k}{4} + t(A)$. Subtraction from the $q^2 + q + 1 - k$ off-arc points proves the identity. Finally,

$$\binom{i-1}{2} \leq \frac{r-2}{r} \binom{i}{2} \quad (3 \leq i \leq r),$$

and the second moment gives the defect bound. $\square$

For six-arcs the defect $t(A)$ is the concurrence count $c(A) = n_3$, and that count is far from arbitrary. Dye’s bound $c(A) \leq 10$ is the top of a seven-value spectrum with a single mechanism behind it.

Proposition 3.2 (Concurrence spectrum). *Let A be a six-arc in $\text{PG}(2, q)$ with q odd, and let $c(A)$ count its nonvertex triple-chord concurrences. Identify the fifteen perfect matchings of A with the fifteen pairs from the six one-factorizations of K_6 , and let G be the graph on those six one-factorizations whose edges are the concurrent matchings, so that $c(A) = |E(G)|$. Then G is the graph of an equivalence relation with at least one odd class. Consequently, if $a_1, \dots, a_r$ are the class sizes, then*

$$c(A) = \sum_i \binom{a_i}{2}, \quad \sum_i a_i = 6, \quad \text{some } a_i \text{ odd},$$

and therefore

$$c(A) \in \{0, 1, 2, 3, 4, 6, 10\}.$$

The value ten occurs exactly for the partition $5 + 1$, in which case the five non-concurrent matchings form a one-factorization.

Proof. Chords through a point off A have pairwise disjoint endpoint pairs, so at most three of them concur and a triple concurrence is carried by a perfect matching. Each of the fifteen perfect matchings lies in exactly two of the six one-factorizations of K_6 , and this incidence is the bijection with pairs used in the statement; disjoint matchings correspond to meeting pairs, and the three matchings containing a fixed edge correspond to a perfect matching of the six one-factorizations.

Two facts about G remain. First, no edge of A lies in three concurrent matchings. This is Dye's edge bound. A matching containing uv pairs the four vertices of $A \setminus \{u, v\}$ in one of three ways, and it is concurrent exactly when the corresponding diagonal point of that complete quadrangle lies on the line uv . All three would put the three diagonal points on uv , and the diagonal points of a quadrangle are noncollinear in odd characteristic [11, §2.2, p. 274]. Each concurrent matching has three edges and each of the fifteen edges of A lies in at most two of them, so $3c(A) \leq 30$. This recovers $c(A) \leq 10$, with equality exactly when the five non-concurrent matchings form a one-factorization. Under the pair dictionary the three matchings through uv are a perfect matching of the six one-factorizations, and every perfect matching there arises this way; so G contains no perfect matching.

Second, adjacency in G is transitive: in a triangle $\{F_i F_j, F_j F_k, F_i F_k\}$ of pairs, if two of the three matchings are concurrent, so is the third.

Such a triangle is exactly a triple whose three matchings span a $K_{3,3}$. The union of three pairwise disjoint perfect matchings of K_6 is a cubic graph on six vertices, hence either $K_{3,3}$ or the triangular prism. The complement of a prism in K_6 is a six-cycle, which splits into two further matchings; a prism triple therefore extends to a one-factorization of K_6 , so its three matchings share a common one-factorization and the triple is a star $\{F_i F_j, F_i F_k, F_i F_l\}$. A triangle triple, whose pairs have no common member, thus spans $K_{3,3}$. Its bipartition splits the six vertices into two triangles, each of the three matchings joins one triangle to the other, and a concurrence of such a matching is a centre of perspective of the two triangles.

Take the common triangle $A_1 = (1:0:0)$, $A_2 = (0:1:0)$, $A_3 = (0:0:1)$ with the first centre of perspective at $(1:1:1)$, which is legitimate because A is an arc, and write the other three vertices as $B_1 = (x:1:1)$, $B_2 = (1:y:1)$, $B_3 = (1:1:z)$. The determinant governing the second perspectivity is $xyz - 1$ and that governing the third is $1 - xyz$, so the two conditions coincide. Only commutativity of the field is used; this determinant identity is the classical theorem that two triangles of a Pappian plane in double perspective are in triple perspective.

A transitive symmetric adjacency is an equivalence relation, so G is a disjoint union of complete graphs; having no perfect matching means some class has odd size. Enumerating the partitions of six with an odd part and evaluating $\sum \binom{a_i}{2}$ gives the displayed set; the all-even partitions 6 and $4 + 2$, which would give 15 and 7, are exactly the ones excluded. The value ten arises only from $5 + 1$. $\square$

Remark 3.3 (What is classical here). The bound $c(A) \leq 10$, its quadrangle mechanism, and the one-factorization structure at equality are Dye's [11, §2.2 and the proof of Theorem 1, pp. 274–275]. The transitivity step is the classical double-implies-triple perspective theorem for a commutative field; what is used here is its reading as a transitivity relation on the six one-factorizations of K_6 . The remaining content is the exclusion of the values five, seven, eight, and nine through that step. The seven-value formulation and the proof given here do not appear in the six-arc sources cited above; Dye's paper treats hexagons with exactly ten Brianchon points. We make no exhaustive absence claim about the broader concurrence-count literature.

The census of the computational companion corroborates the spectrum at $q = 11$: across its fifteen projective classes of six-arcs the count $c(A) = 22 - |\mathcal{U}(A)|$ realizes each of the seven predicted values and none of the excluded ones [21].

4 The rigidity theorem

The conic identification for the displayed arc, proved in Section 6, concerns that one arc. We now prove the symmetry-free characterization stated in Theorem 1.1. The uncovered conic determines the resulting Clebsch hexagon's polarity, and Dye's stabilizer theorem then identifies its projective stabilizer as A_5 ; neither is assumed. The two arguments are independent: the proof of Theorem 1.1 uses only Theorem 3.1, Lemma 4.1 and Dye's results, while Proposition 6.1 uses Theorem 4.4.

Conic containment. The condition is that the uncovered locus $\mathcal{U}(A)$ lies on a conic, not that the arc A itself does. The proof below uses only the former condition and imposes no hypothesis that the six vertices lie on a conic.

We first isolate the geometric input needed to exclude degenerate containing conics.

The exterior-line case is a six-point instance of the nonexistence in odd order of nontrivial *hyperfocused arcs*, whose secants determine only $k - 1$ points on an exterior line [2, 9, 20]. We give a direct proof because the three possible line positions are then handled together.

Lemma 4.1 (Line bound). *Let A be a six-arc in $\text{PG}(2, q)$, where q is odd. Every line contains at most $q - 5$ points of $\mathcal{U}(A)$.*

Proof. A chord contains no point of $\mathcal{U}(A)$. If a non-chord line ℓ contains one vertex, the ten chords on the other five vertices meet ℓ in at least five distinct points. Indeed, two of those chords sharing an endpoint meet only at that endpoint, which lies off ℓ ; their intersections with ℓ therefore give a proper edge-colouring of K_5 , whose chromatic index is five. Together with the vertex, at least six of the $q + 1$ points of ℓ are therefore outside $\mathcal{U}(A)$.

It remains to consider $\ell \cap A = \emptyset$. The fifteen chords meet ℓ in covered points, and at most three chords pass through any one of them: chords sharing an endpoint meet only at that arc vertex, not on ℓ , so all chords through one point of ℓ have disjoint endpoint pairs. Hence there are at least five covered points. Equality would give a proper five-edge-colouring of K_6 , necessarily a one-factorization. The one-factorization of K_6 is unique up to isomorphism [19]; after relabelling, three factors are

$$01|23|45, \quad 02|14|35, \quad 03|15|24.$$

Send ℓ to infinity and normalize the first two directions to horizontal and vertical. Put $p_0 = (0, 0)$, $p_1 = (x, 0)$ and $p_2 = (0, y)$, where $xy \neq 0$. Writing $p_3 = (a, y)$, the first two factors force $p_4 = (x, b)$ and $p_5 = (a, b)$. Parallelism of 03 with 15 and with 24 gives

$$ab - ay + xy = 0, \quad ab - ay - xy = 0.$$

Their difference is $2xy = 0$, impossible for odd q . Thus a disjoint line has at least six covered points, and in all cases at most $(q + 1) - 6 = q - 5$ points of $\mathcal{U}(A)$ remain. $\square$

Proof of Theorem 1.1. Suppose that $\mathcal{U}(A)$ lies on a conic Q . If Q is nonsingular, then $|\mathcal{U}(A)| \leq 12$. If it is degenerate, its rational points lie on at most two rational lines, apart from the nonsplit rank-two case in which only the singular point is rational. Lemma 4.1 again gives $|\mathcal{U}(A)| \leq 12$.

Theorem 3.1, on the other hand, gives

$$|\mathcal{U}(A)| = 22 - c(A),$$

where $c(A)$ counts the nonvertex triple-chord concurrences, namely Dye's Brianchon points. Dye proves $c(A) \leq 10$, with the equality cases, when they occur, forming one $\text{PGL}_3(K)$ -orbit [11, §2.2–2.3, Theorem 1, pp. 275–276]. Here $\text{char } \mathbb{F}_{11} = 11 \neq 2, 5$ and $5 = 4^2$ in $\mathbb{F}_{11}$,

so Dye's equality orbit exists. Thus $|\mathcal{U}(A)| \geq 12$. Both bounds are equalities: $c(A) = 10$, so A is a Clebsch hexagon.

For a Clebsch hexagon the fifteen chords are exterior to Dye's associated nonsingular conic $\mathcal{C}$. Hence $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; both sets have twelve points by the preceding equality, so they coincide. The original Q cannot be degenerate, since Bézout bounds its intersection with $\mathcal{C}$ by four. This proves (i) $\Rightarrow$ (iii), and the same exteriority and count give (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i). Dye's stabilizer theorem identifies the stabilizer with A_5 . $\square$

Corollary 4.2 (Rigidity relative to a fixed conic). *Fix a nonsingular conic $\mathcal{C} \subset \text{PG}(2, 11)$. The six-arcs A with $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_{11})$ form a single orbit under $\text{Stab}_{\text{PGL}(3,11)}(\mathcal{C})$.*

Proof. By Theorem 1.1 any two such arcs are related by a projectivity g , and g carries one uncovered locus to the other, hence $\mathcal{C}(\mathbb{F}_{11})$ to $\mathcal{C}(\mathbb{F}_{11})$. The conic $g(\mathcal{C})$ therefore shares all twelve rational points of $\mathcal{C}$ with $\mathcal{C}$; two distinct conics meet in at most four points by Bézout's theorem, so $g(\mathcal{C}) = \mathcal{C}$ and $g \in \text{Stab}_{\text{PGL}(3,11)}(\mathcal{C})$. $\square$

Projective equivalence of the unordered parity-check column sets is equivalent to monomial equivalence of the associated codes: choose column representatives after the projectivity, then absorb their nonzero scalar factors into coordinate scalings. Thus Theorem 1.1 also proves the coding formulation in the abstract.

Remark 4.3 (Why this rigidity is not in the extension-locus literature). The set $\mathcal{U}(A)$ is the classical extension locus, studied for sufficiently large arcs through Segre's tangent-envelope method; see Ball and Lavrauw [1, Thms. 39–41]. Those results require arc-size hypotheses far from the six-point regime and do not record the conic rigidity detected here.

The same counting argument gives a uniform field window once the uncovered locus fills a conic.

Theorem 4.4 (Conic-filling window and six-arcs). *If $|\mathcal{U}(A)| = q + 1$, then*

$$t(A) = q^2 - mq + m + 3 \binom{k}{4} - k, \quad q^2 - mq + m - k + \frac{6}{r} \binom{k}{4} \leq 0,$$

and $q < m$. Under the same hypothesis $|\mathcal{U}(A)| = q + 1$, if q is even, then $\mathcal{U}(A)$ is not an arc. Hence if $\mathcal{U}(A)$ is a nonsingular conic, then q is odd and

$$2k - 3 \leq q \leq \frac{k(k-1) + 3}{3} < \binom{k}{2}.$$

For $k = 6$, writing $c(A) = n_3$ gives

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \quad c(A) = (q-6)(q-9)$$

when $|\mathcal{U}(A)| = q + 1$, and for q odd Proposition 3.2 restricts the concurrence count to $c(A) \in \{0, 1, 2, 3, 4, 6, 10\}$. For the displayed Clebsch hexagon, Dye's ten Brianchon points give $c(A) = 10$; its fifteen chords have 45 disjoint pairs, cover 115 of the 127 off-arc points, and leave exactly twelve uncovered.

Proof. The first formula is Theorem 3.1 rearranged, and its comparison with the defect bound gives the quadratic inequality. There are $q^2 - k$ covered off-arc points, so $q^2 - k \leq m(q-1)$; if $q \geq m$, then $0 \leq q(q-m) \leq k-m < 0$, a contradiction.

Suppose q is even and $\mathcal{U}(A)$ is an arc. Since it has $q+1$ points, it has a nucleus $N \notin \mathcal{U}(A)$ through which every line is tangent [17, p. 177]: every line through N meets $\mathcal{U}(A)$ in exactly

one point. A chord of A is disjoint from $\mathcal{U}(A)$ and so meets it in no point; hence no chord passes through N . Nor can N be a vertex, since each vertex lies on the $k-1$ chords through it. Thus $N \in \mathcal{U}(A)$, a contradiction.

In the conic case, q is therefore odd and every chord is passant. An off-conic point lies on at most $(q+1)/2$ passants, so the $k-1$ chords through a vertex give $q \geq 2k-3$.

The m chords cover the complement of the conic by passant (elliptic) lines. The exact conic-complement bound of Blokhuis, Brouwer, and Szőnyi gives [3, Proposition 1.6]

$$m \geq 2q-1 - \frac{q+1}{2} = \frac{3(q-1)}{2}.$$

Rearranging gives $q \leq (k(k-1)+3)/3$; the final strict inequality follows from $k \geq 4$.

For $k=6$, one has $r=3$ and $t(A) = n_3 = c(A)$, giving the displayed specialization. For the displayed arc, Dye's Brianchon theorem gives $c(A) = 10$ [11, Theorem 1, pp. 275–276]. At $q=11$, the two moments read $\sum in_i = 150$ and $\sum \binom{i}{2} n_i = 45$, whence $n_1 + n_2 + n_3 = 115$. $\square$

Remark 4.5 (The spectrum alone almost fixes the order). For $k=6$ the two constraints combine. If $\mathcal{U}(A)$ is a nonsingular conic, then q is odd and $c(A) = (q-6)(q-9)$ must lie in $\{0, 1, 2, 3, 4, 6, 10\}$ by Proposition 3.2: the order $q=7$ gives the negative value -2 , every $q \geq 13$ gives more than ten, and the remaining odd orders are $q=5, 9, 11$, with values $4, 0, 10$. The concurrence spectrum therefore recovers a finite list of orders without the conic-complement bound, and the window's lower bound $q \geq 2k-3 = 9$ removes $q=5$. Only $q=9$ then has to be excluded, which the computational companion does with a Sylvester-graph clique bound [21].

Remark 4.6 (Sharpness of the window). Both bounds are attained. A projective four-frame over $\mathbb{F}_5$ has a conic-filling uncovered locus and satisfies $q = 2k-3 = (k(k-1)+3)/3$. The Clebsch six-arc over $\mathbb{F}_{11}$ attains the upper bound. We do not know an infinite family attaining either bound; the computational companion [21] shows that, through $k=8$, these are the only conic-filling examples.

5 The Clebsch family across fields

Recall that A is conic-filling when $\mathcal{U}(A) = \mathcal{Q}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{Q}$. Equivalently, $A \cap \mathcal{Q}(\mathbb{F}_q) = \emptyset$, its chords cover every point outside $A \cup \mathcal{Q}(\mathbb{F}_q)$, and no chord meets $\mathcal{Q}(\mathbb{F}_q)$. For the Clebsch family, the chord-defect identity gives a direct formula for the uncovered locus and isolates the conic-filling specialization.

Proposition 5.1 (The Clebsch-family uncovered formula). *Let K be a Clebsch hexagon over $\mathbb{F}_q$; equivalently, K is a six-arc with ten Brianchon points. By Dye's existence theorem, the characteristic is not two and 5 has a square root in $\mathbb{F}_q$, possibly zero in characteristic five. Then*

$$|\mathcal{U}(K)| = q^2 - 14q + 45.$$

If $q \equiv 3 \pmod{4}$ and $\mathcal{C}$ is Dye's associated conic, then $\mathcal{C}(\mathbb{F}_q) \subseteq \mathcal{U}(K)$ and

$$|\mathcal{U}(K) \setminus \mathcal{C}(\mathbb{F}_q)| = (q-4)(q-11).$$

In particular, among Clebsch hexagons in this congruence class, the associated conic is the complete projective deep-hole syndrome locus if and only if $q=11$.

Proof. Dye defines a Clebsch hexagon by the occurrence of exactly ten Brianchon points, and his Theorem 1 classifies their existence and projective orbit [11, p. 271; Theorem 1, pp. 275–276]. These points are precisely the off-vertex triple-chord concurrences counted by $c(K)$. Thus $c(K) = 10$, and Theorem 3.1 gives

$$|\mathcal{U}(K)| = q^2 - 14q + 55 - 10 = q^2 - 14q + 45.$$

Dye also shows that the edges of K are non-secants of the associated conic when $q \equiv 3 \pmod{4}$ [11, discussion preceding Theorem 6, pp. 281–282]. Hence every rational point of $\mathcal{C}$ is uncovered. Subtracting $|\mathcal{C}(\mathbb{F}_q)| = q + 1$ from the first formula gives

$$q^2 - 15q + 44 = (q - 4)(q - 11).$$

The factor vanishes only at $q = 4$ and $q = 11$, and the stated congruence leaves only $q = 11$. Conversely, Clebsch hexagons exist over $\mathbb{F}_{11}$ by Dye’s Theorem 1, and the inclusion above is then an equality because both sets have twelve points. $\square$

Remark 5.2 (Three Clebsch-family regimes). The polynomial $q^2 - 14q + 45 = (q - 5)(q - 9)$ vanishes exactly at $q = 5, 9$. The root $q = 5$ lies in characteristic five, where the associated-conic construction of Definition 2.1 is not available. At $q = 9$, however, Dye’s construction applies: $5 = -1$ is a square in $\mathbb{F}_9$, and the characteristic is three. Thus the formula gives an empty uncovered locus, so the Clebsch six-arc is complete and not conic-filling. At $q = 11$ the uncovered locus is instead exactly the associated conic.

At $q = 19$, the same icosahedral construction still produces a Clebsch six-arc disjoint from its associated conic, but the covering fails. The twenty conic points remain uncovered [11, p. 281], while Proposition 5.1 gives

$$|\mathcal{U}(A)| = 140 = 20 + 120.$$

Thus $120 = (19 - 4)(19 - 11)$ additional points escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic. The configuration persists; exact conic filling is what the factorization isolates at $q = 11$.

6 The code and its projective syndrome locus

Take the parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 10 & 9 & 4 & 8 & 1 & 1 \\ 0 & 1 & 7 & 5 & 4 & 7 \end{pmatrix},$$

whose columns are the six displayed points of A , and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^\top = 0\}$$

be the resulting $[6, 3, 4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ has determinant $3 \in \mathbb{F}_{11}^\times$), C is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [27].

The arc-coset dictionary from the introduction applies without a GRS hypothesis [8, Def. 6.1, Thm. 6.3]. Each uncovered direction x lifts to $q - 1$ weight-three cosets, and

each has $\binom{6}{3} = 20$ minimum-weight leaders. Geometrically, the same condition says that adjoining x as a seventh column preserves the MDS property.

For example, $s = (1, 0, 0)$ lies on $\mathcal{C}$ and is therefore uncovered. The error

$$e = (7, 4, 1, 0, 0, 0)$$

satisfies $He^T = s$. No error of weight one or two has syndrome s , while the twenty triples of coordinate positions each supports one weight-three leader. Thus this single syndrome exhibits the complete dictionary: its projective direction is a conic point, its coset has distance three, and its nearest-codeword ambiguity has size twenty.

Proposition 6.1 (The uncovered locus is the conic). *The covering radius of C is three, and*

$$\mathcal{U}(A) = \mathcal{D}_{\text{proj}}(C) = \mathcal{C}(\mathbb{F}_{11}).$$

Proof. Theorem 4.4 gives $|\mathcal{U}(A)| = 12$. The exteriority of all fifteen chords gives $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$. Since a nonsingular conic has twelve $\mathbb{F}_{11}$ -points, equality follows without using the orbit classification. The arc-coset dictionary identifies this set with the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three. $\square$

Corollary 6.2 (Directions, cosets, leaders, and received words). *$\mathcal{U}(A)$ has 12 directions, exactly the points of $\mathcal{C}$. Above them lie $12(11 - 1) = 120$ nonzero maximum-distance syndromes, hence 120 deep-hole cosets. Each such coset has $\binom{6}{3} = 20$ minimum-weight leaders, for 2400 leaders in all, and contains $|C| = 11^3 = 1331$ received words. Consequently C has $120 \cdot 1331 = 159720$ received-word deep holes.*

Proof. Immediate from Proposition 6.1, the leader-count clause of the DMP dictionary, and the fact that the fiber of $\mathbb{F}_{11}^3 \setminus \{0\} \rightarrow \text{PG}(2, 11)$ over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has $|C|$ words. $\square$

The stabilizer further organizes all the small point sets that occur in the decoder and orientation constructions.

The next proposition uses a compact two-generator certificate rather than characters or a proper-subgroup census. Independent matrix-to-row checks, block-label invariance, and short reachability words make the seven-orbit decomposition auditable without assuming the orbit labels.

Proposition 6.3 (The A_5 point orbits). *Let $G = \text{Stab}(A) \cong A_5$. Its orbits on $\text{PG}(2, 11)$ have lengths*

$$6, 10, 12, 15, 30, 30, 30.$$

The first four are respectively A , the ten Brianchon points, $\mathcal{C}(\mathbb{F}_{11})$, and the fifteen vertices of the five self-polar triangles. In particular, $\mathcal{C}(\mathbb{F}_{11})$ is the unique twelve-point G -orbit.

Proof. Dye's polarity and stabilizer theorems identify G with the ternary icosahedral representation preserving $\mathcal{C}$, and his Corollary 1 makes it transitive on the six vertices, ten Brianchon points, and fifteen triangle vertices [11, Theorems 2–3 and Corollary 1, pp. 276–279]. For completeness, the orbit lengths themselves have a shorter finite certificate that avoids characters and the proper-subgroup census. Choose the normalized projectivities

$$s = \begin{pmatrix} 7 & 6 & 6 \\ 8 & 9 & 8 \\ 6 & 6 & 7 \end{pmatrix}, \quad t = \begin{pmatrix} 7 & 6 & 6 \\ 6 & 8 & 3 \\ 2 & 1 & 7 \end{pmatrix} \quad \text{in } \text{PGL}_3(11).$$

On the six vertices they induce

$$s = (3\ 4)(5\ 6), \quad t = (1\ 2\ 3)(4\ 6\ 5).$$

A breadth-first word ledger in s, t recovers all sixty distinct support permutations; its longest word has length twelve. Thus s, t generate the displayed order-sixty action.

In the normalization by first nonzero coordinate, the seven orbit seeds and their certified sizes are

seed	size	geometric block
[1:0:0]	12	associated conic
[1:0:1]	30	—
[1:0:2]	30	—
[1:0:3]	10	Brianchon points
[1:0:4]	15	self-polar triangle vertices
[1:0:6]	30	—
[1:1:7]	6	hexagon vertices.

The two 133-entry action rows computed from the displayed matrices preserve these blocks. Starting at each seed, a second breadth-first ledger reaches every point of its block, with longest word length nine. Hence preservation gives containment of each generated orbit in its labelled block, while the word ledger gives the reverse containment. The matrix-to-row checks are independent of the block labels, so this is an action certificate rather than an orbit table used as its own justification. The formal terminal `twoGenerator_pointOrbit_partition` checks both ledgers, the label invariance, the partition, and these cardinalities by ordinary kernel reduction. Independently, the paper-owned exact replay `check_code_automorphisms.py` reconstructs all 133 points from the displayed matrices and seeds, verifies the two support permutations, and recomputes both breadth-first depth bounds and the seven-orbit partition.

The six-, ten-, and fifteen-point blocks are the transitive sets identified by Dye above. His Theorem 6(v), specialized at $q = 11 \equiv 1 \pmod{10}$, identifies the twelve-point block with $\mathcal{C}(\mathbb{F}_{11})$ [11, pp. 281–282; §3.1, p. 284]; since the block ledger has only one twelve-point member, this orbit is unique. $\square$

6.1 Automorphisms and deep-hole orbits

The projective stabilizer of the six column rays is A_5 of order 60, acting faithfully and 2-transitively on the coordinate supports. It is the support-permutation quotient of the monomial automorphism group:

$$1 \longrightarrow \mathbb{F}_{11}^\times \longrightarrow \text{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$

The kernel consists of the ten global coordinate scalars: a projectivity fixing all six column rays fixes a projective frame and is the identity. The sequence splits. Since $\gcd(3, 10) = 1$, each projective class has a unique determinant-one representative. These representatives are closed under multiplication, and their induced monomial lifts are unique once the displayed column representatives are fixed; hence they give an A_5 complement. The global scalars are central, so

$$\text{MAut}(C) \cong C_{10} \times A_5$$

has order 600.

Proposition 6.4 (One orbit on received-word deep holes). *The monomial automorphism group of C is transitive on the 120 deep-hole cosets. Let Γ be the group generated by these automorphisms and by translations $v \mapsto v + c$ for $c \in C$. Then*

$$|\Gamma| = |C| |\text{MAut}(C)| = 1331 \cdot 600 = 798600,$$

and Γ is transitive on all 159720 received-word deep holes. The stabilizer of each such word is cyclic of order five.

Proof. The projective stabilizer A_5 is transitive on the twelve points of $\mathcal{C}$. Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six fixed-point chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust $\mathcal{C}(\mathbb{F}_{11})$. Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above $\mathcal{C}$. Its central scalar subgroup $\mathbb{F}_{11}^\times$ acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all $12 \cdot 10 = 120$ maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes v and w , choose $m \in \text{MAut}(C)$ with $\sigma(mv) = \sigma(w)$. Then $c = w - mv$ has zero syndrome, so $c \in C$, and the codeword translation by c sends mv to w . Both kinds of generator are Hamming isometries preserving C . The monomial group normalizes the translation subgroup, since $m t_c m^{-1} = t_{m(c)}$ with $m(c) \in C$. Thus $\Gamma = C \rtimes \text{MAut}(C)$; the two subgroups have trivial intersection and the displayed product order follows. Orbit–stabilizer now gives stabilizer order $798600/159720 = 5$, and every group of prime order is cyclic. $\square$

6.2 Decoding consequences

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight 0, 1, 2, 3, summing to $11^3 = 1331$); the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters. This is the radius-three instance of the coset-weight framework of Davydov, Marcugini, and Pambianco [8, §6]; the secant indices below refine its weight-two stratum by nearest-codeword multiplicity.

The special geometry also makes distance diagnosis and nearest-codeword structure unusually explicit. Write a syndrome as $s = (s_0, s_1, s_2)$ and put $Q(s) = s_0 s_2 - s_1^2$. For a point $x \notin A$, its *secant index* is the number of chords of A through x ; by the dictionary above, this is the number of weight-two leaders for each nonzero syndrome on the ray x . The following oracle is likewise derived from incidence identities; the syndrome enumeration in the formal package is an independent check of it, not an input to its proof.

Proposition 6.5 (Complete syndrome oracle and ambiguity enumerator). *For $v \in \mathbb{F}_{11}^6$ with syndrome $s = Hv^\top$,*

$$d(v, C) = \begin{cases} 0, & s = 0, \\ 1, & [s] \in A, \\ 3, & s \neq 0 \text{ and } Q(s) = 0, \\ 2, & \text{otherwise.} \end{cases}$$

Among the 1330 nonzero cosets, the numbers having respectively one, two, three, and twenty nearest codewords are

$$960, \quad 150, \quad 100, \quad 120.$$

For a distance-two syndrome the number of nearest codewords is the secant index of its direction. Every distance-three syndrome has exactly one leader on each of the twenty three-element coordinate supports.

Proof. The DMP dictionary identifies weight-one directions with the six columns, weight at most two with the secant union, and weight three with its complement. Proposition 6.1 identifies the last set with $Q = 0$, proving the oracle. For completeness, the ambiguity counts come from three incidence equations rather than a syndrome table. If n_i counts off-arc points of secant index i , then

$$n_0 = 12, \quad n_1 + n_2 + n_3 = 133 - 6 - 12 = 115,$$

while chord-point and disjoint-chord-pair incidence give

$$n_1 + 2n_2 + 3n_3 = 15(11 - 1) = 150, \quad n_2 + 3n_3 = 3 \binom{6}{4} = 45.$$

Dye's ten Brianchon points give $n_3 = 10$, and hence $(n_0, n_1, n_2, n_3) = (12, 90, 15, 10)$. Every direction has ten nonzero affine syndromes. Thus the distance-two cosets contribute 900 cases of multiplicity one, 150 of multiplicity two, and 100 of multiplicity three; the sixty weight-one cosets add sixty more cases of multiplicity one. Leaders are in bijection with nearest codewords by translation within the coset. Finally, the three columns on any support are independent. For an uncovered syndrome their unique coefficients are all nonzero, since a zero coefficient would put the syndrome on a secant. Hence all twenty supports, and no others, give its weight-three leaders. $\square$

6.3 Self-polar structure

Edge's finite-plane construction [12, §§30.2–31] and Dye's general-field synthetic theory [11, Theorem 2, pp. 277–278] exhibit five triangles on the six vertices, self-polar for $\mathcal{C}$. Their vertex-pair matchings partition the fifteen pairs into five perfect matchings of K_6 . In the classical terminology this one-factorization is a *synthematic total*; the action of S_6 on the six totals is the standard construction of its outer automorphism [13]. Section 7 turns its ten pairs of matchings into the Brianchon-support dictionary.

7 The invariant support bipartition

Label the six coordinates by $0, \dots, 5$. The 20 three-element coordinate supports form two complementary A_5 -orbits of size ten. Automorphisms preserve the two classes, but nothing intrinsic chooses one as positive. We use *support bipartition* only as shorthand for this unoriented bipartition. Its ten complementary pairs carry the Petersen graph in Figure 2.

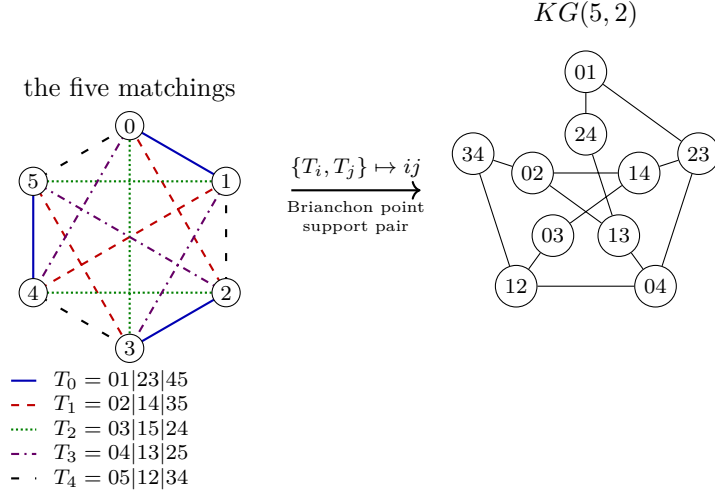


Figure 2: The five-matchings–Petersen dictionary. The line styles encode the five self-polar triangles as perfect matchings of the six coordinates. For each pair $\{T_i, T_j\}$, their union is an alternating six-cycle: its opposite-edge matching gives a Brianchon point, and its alternating vertex classes give a complementary support pair. Both are represented by the Petersen vertex ij ; adjacency means that the index pairs are disjoint.

Proposition 7.1 (Brianchon–support dictionary). *Let $\mathcal{T}$ be the five self-polar triangles of Edge and Dye [12, 11], regarded as five perfect matchings, and let $\mathcal{B}(A)$ be the set of ten Brianchon points of the Clebsch hexagon. For distinct $T, T' \in \mathcal{T}$, the union $T \cup T'$ is an alternating six-cycle. Let $M(T, T')$ be the perfect matching joining opposite vertices of this cycle. Let $\beta(T, T') = \{S, S^c\}$ be the unordered pair of alternating vertex classes of the cycle. For example, $T_0 \cup T_1$ is the cycle $0, 1, 4, 5, 3, 2$, so $M(T_0, T_1) = 05|13|24$ and $\beta(T_0, T_1) = \{\{0, 3, 4\}, \{1, 2, 5\}\}$.*

The three chords indexed by $M(T, T')$ concur at a point $b(T, T') \in \mathcal{B}(A)$. The maps

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \quad \{T, T'\} \longmapsto b(T, T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S, S^c\} : |S| = 3\}, \quad \{T, T'\} \longmapsto \beta(T, T'),$$

are A_5 -equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of $\mathcal{T}$.

Proof. The opposite-vertex matching is disjoint from T and T' and contains one edge from each of the other three members of $\mathcal{T}$. Hence $\{T, T'\}$ is recovered as the two members of $\mathcal{T}$ disjoint from $M(T, T')$, so the ten triangle pairs give exactly the ten perfect matchings outside $\mathcal{T}$. Applying the alternating-class rule to the five displayed matchings gives ten distinct complementary support pairs, hence all ten such pairs. Edge’s Clebsch construction says that the three chords of each such matching concur at the ten distinct Brianchon points [12, §§19–20, 30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above.

The ten outside matchings give the ten triple concurrences. The remaining fifteen intersections of disjoint chords have multiplicity one. Since every step uses only the five

invariant matchings and incidence, both bijections and their compatibility with the support map are A_5 -equivariant. $\square$

Corollary 7.2 (The decoder reconstructs the Brianchon configuration). *The ten projective directions having three nearest weight-two errors are exactly the ten Brianchon points. At each such direction the three error supports are pairwise disjoint and hence form a perfect matching of the six coordinates. The ten matchings thereby recovered are precisely the ten perfect matchings outside the five matchings in $\mathcal{T}$.*

Proof. The three weight-two representations of a secant-index-three direction use the three chords through it. Proposition 7.1 identifies the ten triple concurrences with the ten Brianchon points and their chords with exactly the ten matchings outside $\mathcal{T}$. $\square$

Proposition 7.3 (Invariant support bipartition). *Only the bipartition is intrinsic: neither part has a canonical sign. The twenty three-element coordinate supports split into two complementary A_5 -orbits $\mathcal{O}_+$ and $\mathcal{O}_-$ of size ten. For every maximum-distance syndrome s , each support determines exactly one weight-three leader of the coset $H^{-1}(s)$; hence that coset has ten leaders supported in each class. Globally, the 2400 minimum-weight leaders split as $1200 + 1200$.*

Every monomial automorphism of C preserves this unordered bipartition. The nontrivial coset of A_5 in the exotic normalizer $N_{S_6}(A_5) \cong S_5$ exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. By Proposition 7.1, the ten complementary support pairs are canonically identified both with the two-subsets of the five self-polar triangles and with the ten Brianchon points.

Proof. The degree-six action of A_5 on the twenty three-subsets has two ten-element orbits, exchanged by complementation. It preserves a unique set of five perfect matchings, also preserved by its full S_5 normalizer. The alternating-cycle construction of Figure 2 is therefore equivariant, and the outside normalizer coset exchanges the two support orbits. Fix a concrete finite witness by

	permutation of $\{0, \dots, 5\}$	action on $\{T_0, \dots, T_4\}$	action on $\{\mathcal{O}_+, \mathcal{O}_-\}$	
s	$(2\ 3)(4\ 5)$	$(T_1\ T_2)(T_3\ T_4)$	preserves each	(*)
t	$(0\ 1\ 2)(3\ 5\ 4)$	$(T_0\ T_4\ T_1)$	preserves each	
r	$(2\ 4\ 3\ 5)$	$(T_1\ T_3\ T_2\ T_4)$	exchanges them.	

Applying these permutations to the five displayed matchings verifies the middle column directly; applying them to the twenty three-subsets verifies the last. The first two generators generate the order-60 subgroup A_5 , while all three generate its order-120 normalizer. Thus the table supplies both preservation and exchange without an undisplayed action list. Fix a maximum-distance syndrome s and a three-support S . The three columns indexed by S are independent because A is an arc, so they give a unique coefficient vector representing s . None of its three coefficients can vanish: otherwise $[s]$ would lie on a secant of A , contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support S . Finally, every monomial automorphism induces a support permutation in A_5 , by the exact sequence in Section 6.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits but cannot preserve C , even after coordinate rescaling, because its support permutations lie outside that quotient A_5 . $\square$

8 The intrinsic orientation two-graph

The support bipartition and the twelve deep-hole directions carry two apparently different orientation choices. We now show that they are the same datum. This comparison uses

only the reconstructed A_5 -sets in this paper. We use the standard correspondence between two-graphs, switching classes, and Seidel matrices. Two-graphs are due to Higman, with Taylor the standard reference and Seidel's switching the graph-side description; see Brouwer and Van Maldeghem [31, §1.1.12] for the correspondence with attributions, and [10, §2] for the presentation used here.

The cubic threefold that appears below agrees with a known six-nodal model. Cheltsov, Tschinkel, and Zhang classify cubics already assumed to have six nodes in general linear position and, in the S_5 case, give the equation [5, §7, equations (7.4)–(7.5), Proposition 7.3]. Choosing $x_0 = 0$ as a representative of $\mathbf{Q}^6/\mathbf{Q}^1$ and interchanging their coordinates y_3 and y_5 identifies their equation (7.5) with $C = 0$ below. More explicitly, if $G(y_1, \dots, y_5)$ denotes the left-hand side of their equation (7.5), then

$$C(0, y_1, y_2, y_3, y_4, y_5) = G(y_1, y_2, y_5, y_4, y_3).$$

The result here is not a new construction of that cubic or its abstract S_5 -action. It recovers the cubic intrinsically from the syndrome locus: the support-orbit signs give its equation, and the diagonal determinant identity identifies it with the golden continuation operator. Because the cited proposition assumes six nodes, we prove singular-locus completeness independently below; the coordinate match is not used in that step.

Let $\Omega = \mathcal{C}(\mathbb{F}_{11})$. Its points have stabilizer C_5 . The two points fixed by each Sylow 5-subgroup form an antipodal pair, giving an A_5 -equivariant quotient

$$\Omega = A_5/C_5 \longrightarrow A_5/D_5 = \Xi$$

onto six axes. Write R for its deck involution. The stabilizer of a point of Ω has suborbits of lengths 1, 1, 5, 5; let A and A' be the adjacency matrices of the two five-valent orbitals.

Theorem 8.1 (Support cubic and golden continuation). *The syndrome locus reconstructs an unordered orientation torsor with the following equivalent presentations.*

- (i) *Give the two A_5 -orbits of three-coordinate supports opposite signs. Their signed moments in degrees zero, one, and two vanish, while the first nonzero moment is the cubic line spanned by*

$$C(x) = \sum_{i < j < k} c_{ijk} x_i x_j x_k, \quad c_{ijk} \in \{\pm 1\}.$$

- (ii) *On the fibre-odd lattice*

$$L^- = \{f \in \mathbf{Z}^\Omega : f(R\omega) = -f(\omega)\},$$

either five-valent orbital induces, after choosing one representative over each axis, a symmetric signed matrix B with

$$B_{ii} = 0, \quad B_{ij} \in \{\pm 1\}, \quad B^2 = 5I.$$

Changing the six representatives switches B to DBD , and exchanging the orbitals negates it.

The two presentations determine one another. Under the unique A_5 -equivariant identification of the coordinate positions with Ξ ,

$$\boxed{c_{ijk} = B_{ij} B_{jk} B_{ki}.$$

Equivalently, if $e_d(x)$ denotes the elementary symmetric polynomial, then

$$\boxed{\det(B + \text{diag}(x_0, \dots, x_5)) = e_6(x) - e_4(x) + 5e_2(x) - 125 - 2C(x).}$$

Thus the support cubic is the sole nonsymmetric term in the diagonal determinant pencil of the golden operator. In homogeneous form, with $F_B(x, z) = \det(zB + \text{diag}(x))$, this becomes

$$F_B(x, z) - F_B(x, -z) = -4z^3C(x).$$

The cubic descends to the augmentation space $\mathbf{Q}^6/\mathbf{Q}\mathbf{1}$. Conversely, its two-graph identities and vanishing signed pair moments reconstruct B and force $B^2 = 5I$. On six labelled axes there are exactly twelve such switching classes. The group S_6 permutes them transitively with point stabilizer of order sixty, and negation acts on them without fixed points, so the six unordered pairs $\{[B], [-B]\}$ carry a transitive S_6 -action with point stabilizer of order 120.

Corollary 8.2 (Nodes, symmetry, and integral commutant). *The cubic also recovers the unlabelled axis carrier: in*

$$\mathbf{P}(\mathbf{Q}^6/\mathbf{Q}\mathbf{1})$$

its zero locus has exactly six singular points $p_a = [\mathbf{1} - 6e_a]$, all ordinary nodes. They form a projective frame, and

$$\text{Aut}_{\text{proj}}(C = 0) \cong S_5,$$

with the oriented-cubic stabilizer A_5 as its index-two subgroup. Consequently the continuation algebra and its integral commutant are

$$\mathbf{Q}[B] \cong \mathbf{Q}(\sqrt{5}), \quad \text{End}_{\mathbf{Z}A_5}(L^-) = \mathbf{Z}[B] \cong \mathbf{Z}[\sqrt{5}].$$

Proof of Theorem 8.1 and Corollary 8.2. The proof has five stages: the orbital pentagon produces the golden operator; triangle holonomy identifies its switching class with the support cubic; principal minors give the determinant pencil; the singular locus recovers the node frame and its symmetry; and Galois descent gives the rational and integral commutants.

The orbital pentagon. The degree-six action has point stabilizer D_5 . Its two orbits on three-subsets have size ten, are exchanged by complementation, and produce a nonzero signed cubic determined up to its overall sign.

Antipodal exchange carries one five-suborbit of Ω to the other, so $A' = AR$. Both orbitals are self-paired. Indeed, in the coset model $\Omega = A_5/H$ with $H = \langle r \rangle$, $r = (12345)$, the two nontrivial double cosets have representatives $a = (345)$ and $b = (23)(45)$; they are inverse-stable because $r^2ar^2 = a^{-1}$ and $b = b^{-1}$.

The stabilizer H of a point acts regularly on the other five axes. Each five-suborbit is an H -orbit, so its projection to those axes is an equivariant map between regular five-sets and hence a bijection. Thus between two fibres exactly one point lies in $A(\omega)$ and the other in $A'(\omega)$. On L^- the two orbitals act oppositely. Choosing one point above each axis therefore represents the restriction of A by a symmetric zero-diagonal matrix B with off-diagonal entries ± 1 . Changing a lift switches the corresponding row and column; exchanging the orbitals negates B .

Fix ω and choose the other five lifts in $A(\omega)$, so $B_{0i} = 1$. The stabilizer C_5 of ω acts regularly on the other five axes. Their signs are therefore constant on the sides and on the diagonals of a pentagon. These signs cannot agree. Indeed, the five-valent orbital graph is

connected: a representative of the five-point suborbit lies outside D_5 , and D_5 is the only proper subgroup of A_5 strictly containing C_5 . Equal positive signs would make the graph two disjoint copies of K_6 . Equal negative signs would give, after switching, $K_{6,6}$ minus the antipodal matching. Its unique bipartition would be preserved by A_5 , which has no quotient of order two, contradicting transitivity on Ω .

Thus side and diagonal signs are opposite. After relabelling the pentagon and, if necessary, exchanging the two orbitals, we obtain the following gauge.

Triangle holonomy. For concreteness, one such gauge is

$$B = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & -1 & -1 \\ 1 & 1 & 0 & -1 & 1 & -1 \\ 1 & 1 & -1 & 0 & -1 & 1 \\ 1 & -1 & 1 & -1 & 0 & 1 \\ 1 & -1 & -1 & 1 & 1 & 0 \end{pmatrix}.$$

This is a symmetric conference matrix of order six. Up to switching and permutation, every conference matrix of order six is the Paley matrix [15, §2]. The classical classification supplies context for the gauge; the intrinsic recovery from syndrome and support data is the content here. The pentagon gives $B^2 = 5I$ directly. The dot product of row zero with any other row is zero because the four remaining entries split as two plus and two minus. For two nonzero rows, the contribution through vertex zero is 1, while the three remaining pentagon vertices contribute -1 , whether the two indices are adjacent or diagonal. Thus every off-diagonal entry of B^2 vanishes, while each diagonal entry is five. Since $A - A'$ vanishes on the fibre-even lattice and acts as $2B$ on L^- , this also gives the full orbital identity

$$(A - A')^2 = 10(I - R).$$

It also recovers the original intersection count. For a nonantipodal pair, two of the four remaining axes have equal signs and two have opposite signs; each axis has two lifts. Hence the eight intermediate points contribute four positive and four negative products.

Its triangle products are switching-invariant. They are constant on the two A_5 -orbits of three-subsets, take opposite values on complementary triples, and are not constant; after choosing the overall sign of C they are exactly the coefficients c_{ijk} . For example,

$$c_{012} = B_{01}B_{12}B_{20} = 1, \quad c_{345} = B_{34}B_{45}B_{53} = -1.$$

These triples are complementary, so A_5 -invariance proves the sign rule on both orbits. Conversely, the coefficients satisfy

$$c_{ijk}c_{ij\ell}c_{ik\ell}c_{j\ell k} = 1 \quad (i < j < k < \ell).$$

Choose a temporary gauge with $B_{0i} = 1$. Then $B_{ij} = c_{0ij}$, and the four-point identity recovers all remaining triangle products. Thus the cubic recovers the switching class of B . Moreover, for $i \neq j$,

$$\sum_{k \neq i, j} c_{ijk} = B_{ij} \sum_k B_{ik}B_{kj} = B_{ij}(B^2)_{ij} = 0.$$

Summing these equations gives the signed one-index and total sums as zero. These are exactly the signed moments below degree three, and they imply $C(x + t\mathbf{1}) = C(x)$.

The implication reverses. Once the four-point identities have reconstructed the switching class,

$$(B^2)_{ij} = B_{ij} \sum_{k \neq i, j} c_{ijk} \quad (i \neq j),$$

while every diagonal entry of B^2 is five. Thus signed pair balance is equivalent to $B^2 = 5I$.

The gauge $B_{0i} = 1$ is reached by switching with $D_0 = 1$ and $D_i = B_{0i}$, and it is the unique such representative of its switching class, since D is determined once D_0 is fixed and D and $-D$ act alike. In this gauge the equation $(B^2)_{0i} = 0$ reads $\sum_{k \neq 0, i} B_{ki} = 0$: four signs summing to zero, so exactly two are positive. The positive edges on $\{1, \dots, 5\}$ therefore form a 2-regular graph, hence a pentagon.

The converse also holds, and it is the direction the later argument uses. Let P be any pentagon on $\{1, \dots, 5\}$ and let B be the matrix its gauge defines. Every diagonal entry of B^2 is five. For nonzero $i \neq j$, the term $k = 0$ contributes $B_{i0}B_{0j} = 1$, while each of the three remaining indices contributes $+1$ exactly when it is P -adjacent to both or to neither of i and j . If i and j are P -adjacent, one of the three is adjacent to neither and two are adjacent to exactly one; if they are not, one is adjacent to both and two to exactly one. Either way the three terms are $+1, -1, -1$, so $(B^2)_{ij} = 1 - 1 = 0$ and B is balanced.

Gauge classes at 0 therefore correspond bijectively to the $5!/(5 \cdot 2) = 12$ pentagons on the other five axes. The stabilizer of 0 in S_6 preserves the gauge and acts transitively on those pentagons with dihedral stabilizer of order ten, so S_6 is transitive on the twelve classes and every point stabilizer has order $720/12 = 60$. To compare $-B$ with the chosen gauge, first negate B and then switch by $D_0 = 1$, $D_i = -1$; this restores $B_{0i} = 1$ and replaces the pentagon by its pentagram. Global negation itself is not switching, and it fixes no switching class; the twelve classes pair into six with stabilizers of order 120. Since triangle products are switching-invariant, the order-sixty stabilizer preserves every c_{ijk} while the outer coset reverses them all; this is the $A_5 \subset S_5$ distinction of Corollary 8.2, seen from the two-graph side.

The determinant pencil. For a triple $S = \{i, j, k\}$, its principal minor is

$$\det B_S = 2B_{ij}B_{jk}B_{ki} = 2c_{ijk}.$$

The identity $B^2 = 5I$ gives $B^{-1} = B/5$ and $\det B = -125$. Jacobi's complementary-minor identity then gives principal-minor values 5 in size four and 0 in size five, while complementary size-three minors have opposite signs. Expanding along the diagonal variables gives the displayed determinant formula and its homogeneous odd-part identity. In particular, the same quadratic identity that defines the golden operator also recovers support complementation.

The node frame. It remains to see that the cubic itself recovers the six axes. We use the determinantal duality behind the golden operator. Put $E = \mathbf{Q}(\sqrt{5})$, $t_{\pm} = (1 \pm \sqrt{5})/2$, and

$$U(t) = \begin{pmatrix} t & t & -1 \\ t & 1 & -t \\ 1 & t & -t \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The displayed conference identity gives $BU(t_{\pm}) = \pm\sqrt{5}U(t_{\pm})$ and $U(t_-)^{\top}U(t_+) = 0$; write $V_{\pm}$ for the respective column spaces. Thus, for $D_x = \text{diag}(x_0, \dots, x_5)$,

$$\Phi_x = U(t_-)^{\top}D_xU(t_+)$$

is a matrix for the cross-golden block and is unchanged by $x \mapsto x + t\mathbf{1}$. Direct expansion of this 3×3 determinant gives the exact identity

$$\det(\Phi_x) = -C(x).$$

The nonzero A_5 -map $x \mapsto \Phi_x$ has irreducible source V_5 , so it identifies the augmentation module with the five-dimensional summand in

$$\text{Hom}_E(V_+, V_-) \cong V_4 \oplus V_5;$$

this is the standard tensor-product rule $V_3 \otimes V_{3'} = V_4 \oplus V_5$. Using the invariant forms, regard $\Phi \in \text{Hom}_E(V_+, V_-)$ and $\Psi \in \text{Hom}_E(V_-, V_+)$. Pair these spaces by the invariant perfect pairing

$$\langle \Psi, \Phi \rangle = \text{tr}(\Psi\Phi),$$

Its annihilator has dimension four and is the complementary V_4 , denoted Λ . In the displayed eigenbases, solving the six coefficient equations $\text{tr}(\Psi\Phi_x) = 0$ gives the four-dimensional solution space

$$\Psi_z = \begin{pmatrix} 0 & a(z) & b(z) \\ z_0 & 0 & z_1 \\ z_2 & z_3 & 0 \end{pmatrix}, \quad z = (z_0, z_1, z_2, z_3),$$

where

$$\begin{aligned} a(z) &= (\sqrt{5} - 1)z_0 + \frac{3 - \sqrt{5}}{2}z_1 + \frac{\sqrt{5} - 5}{2}z_2 + (\sqrt{5} - 1)z_3, \\ b(z) &= \frac{\sqrt{5} - 5}{2}z_0 + (1 - \sqrt{5})z_1 + (\sqrt{5} - 1)z_2 + \frac{\sqrt{5} - 3}{2}z_3. \end{aligned}$$

Thus this family is all of Λ , and

$$\det(\Psi_z) = a(z)z_1z_2 + b(z)z_0z_3.$$

At $z = (1, 0, 0, 1)$ the determinant is $\sqrt{5} - 4 \neq 0$. Hence determinant restricts to a nonzero A_5 -invariant cubic on Λ . The character $\chi_{V_4} = (4, 0, 1, -1, -1)$ on the classes $1, 2, 3, 5A, 5B$ gives the symmetric-cube character $(20, 0, 2, 0, 0)$, hence

$$\dim(\text{Sym}^3 V_4^*)^{A_5} = \frac{20 + 20 \cdot 2}{60} = 1.$$

Therefore this determinant is, up to scalar, the Clebsch diagonal cubic

$$z_1^3 + \dots + z_5^3 = 0, \quad z_1 + \dots + z_5 = 0.$$

This cubic surface is smooth: at a singular point the five squares z_i^2 would be equal, but a sum of five signs cannot vanish in characteristic zero.

Completeness of the singular locus needs neither that smoothness nor a transversality hypothesis. Because $\det(\Phi_x) = -C(x)$, differentiating the cross-golden determinant along a centered coordinate line and evaluating identifies each of its five partial derivatives with the corresponding gradient quadric of C , up to sign. Here is an exact exhaustion of their

common zero set. Represent the translation quotient by $x_0 = 0$ and write $y_i = x_i$ for $1 \leq i \leq 5$. In these coordinates the five gradient quadrics are

$$\begin{aligned} q_1 &= -y_2y_3 - y_2y_4 + y_2y_5 + y_3y_4 - y_3y_5 + y_4y_5, \\ q_2 &= -y_1y_3 - y_1y_4 + y_1y_5 + y_3y_4 + y_3y_5 - y_4y_5, \\ q_3 &= -y_1y_2 + y_1y_4 - y_1y_5 + y_2y_4 + y_2y_5 - y_4y_5, \\ q_4 &= -y_1y_2 + y_1y_3 + y_1y_5 + y_2y_3 - y_2y_5 - y_3y_5, \\ q_5 &= y_1y_2 - y_1y_3 + y_1y_4 + y_2y_3 - y_2y_4 - y_3y_4. \end{aligned}$$

Stratify projective four-space by the largest nonzero coordinate: in chart r , set $y_r = 1$ and $y_{r+1} = \cdots = y_5 = 0$. With lexicographic order, the reduced Gröbner bases of the five gradient quadrics are

r	reduced basis in $\mathbf{Q}[y_1, \dots, y_{r-1}]$
1	$\emptyset$
2	y_1
3	y_1, y_2
4	y_1, y_2, y_3
5	$y_1 - y_4, y_2 - y_4, y_3 - y_4, y_4(y_4 - 1).$

The first four charts give the first four coordinate classes; the last gives the fifth coordinate class and the all-ones class. In the translation quotient these are exactly the six centered axis classes $[\mathbf{1} - 6e_a]$. Hence the determinantal cubic threefold in $\mathbf{P}(\Lambda^\perp)$ has exactly the six classes $p_a = [\mathbf{1} - 6e_a]$ as singular points, and these already form a projective frame because the kernel of the cross-golden compression is the all-ones line. The same derivative identity and chart classification are also kernel-checked: the derivative identity over any commutative ring containing a root of $t^2 = t + 1$ and the classification of singular points over any field of characteristic zero containing such a root; Section 9 names the declarations.

The same pairing appears in the literature: the cross-golden four-space Λ and its trace orthogonal $\Lambda^\perp$ carry the paired determinantal hypersurfaces of Hassett and Tschinkel, whose Proposition 10 characterizes the trace-dual partner of a smooth determinantal cubic surface as a cubic threefold with six ordinary double points in linear general position [16, §3, Proposition 10]. That equivalence is context here; the argument above does not use it.

The node type follows directly from the same golden identity. At p_a , pair balance gives

$$\text{Hess}(C)_{ij}(p_a) = \begin{cases} -6c_{aij}, & a \notin \{i, j\}, \\ 0, & a \in \{i, j\}. \end{cases}$$

The nonzero block is switching-equivalent to a five-by-five principal minor M of B . If u is the deleted row, the corresponding block decomposition of $B^2 = 5I$ gives

$$Mu = 0, \quad M^2 = 5I - uu^\top.$$

Since $u^\top u = 5$, the second identity restricts to $M^2 = 5I$ on $u^\perp$. Thus $\ker M = \mathbf{Q}u$ and $\text{rank } M = 4$. Moreover $\text{tr } M = 0$, so the two square roots of 5 occur twice each and

$$\chi_M(\lambda) = \lambda(\lambda^2 - 5)^2.$$

The Hessian therefore has precisely the translation and radial directions as its kernels. Its induced form on the four-dimensional projective tangent space is nondegenerate, so every singularity is an ordinary double point.

Frame symmetry. The node frame identifies every projective cubic automorphism with a frame permutation. Let $N \leq S_6$ normalize the five matchings $\mathcal{T}$ in Figure 2. Its action on $\mathcal{T}$ is faithful, and the finite table $(*)$ supplies normalizers inducing generators of A_5 and an odd permutation; hence $N \cong S_5$. The even generators preserve every c_{ijk} and the odd element reverses them all. Moreover, the identity and the five transpositions $(0\ i)$ give six distinct cubic lines. Thus the line stabilizer has order at most $720/6 = 120$ and equals N ; its orientation character is the sign on $\mathcal{T}$. The cubic-line and oriented-cubic stabilizers are therefore S_5 and A_5 .

The integral commutant. Finally, after extension from $\mathbf{Q}$ to $E = \mathbf{Q}(\sqrt{5})$, the odd module splits as the two nonisomorphic three-dimensional icosahedral representations:

$$L^- \otimes_{\mathbf{Z}} E \cong V_3 \oplus V_{3'}.$$

They are exchanged by the nontrivial element of $\text{Gal}(E/\mathbf{Q})$, and B acts on them by the conjugate scalars $\sqrt{5}$ and $-\sqrt{5}$. Schur's lemma followed by Galois descent therefore gives

$$\text{End}_{\mathbf{Q}A_5}(L^- \otimes \mathbf{Q}) = \mathbf{Q}[B] \cong E.$$

If $aI + bB$ preserves L^- , its diagonal entries force $a \in \mathbf{Z}$, while any off-diagonal unit forces $b \in \mathbf{Z}$. This proves the integral commutant statement. $\square$

Remark 8.3 (The conductor at two). The order $\mathbf{Z}[\sqrt{5}]$ has conductor two in $\mathbf{Z}[(1 + \sqrt{5})/2]$. The orientation collapse is already visible without normalizing: modulo 2, all c_{ijk} become 1, while $B - I$ has rank one and square zero. This is precisely the degeneration

$$\mathbf{Z}[\sqrt{5}] \otimes \mathbb{F}_2 \cong \mathbb{F}_2[u]/(u - 1)^2.$$

No ambient characteristic-zero geometry is used here.

9 Proof sources and formal cross-check

The rigidity and field-window arguments use three external mathematical inputs, while the orientation section uses one external model identification and no external singular-locus theorem. Dye's Brianchon bound and equality classification identify the Clebsch orbit [11, §2.2–2.3, Theorem 1, pp. 275–276]; his stabilizer and associated-conic results supply the classical Clebsch geometry [11, Theorems 2–3, pp. 276–278]. The even-order oval obstruction uses the standard nucleus theorem [17, p. 177]. The upper conic-filling bound is Proposition 1.6 of Blokhuis, Brouwer, and Szőnyi [3]. Cheltsov, Tschinkel, and Zhang supply a published S_5 -symmetric equation inside their classification of cubics already assumed to be six-nodal. Section 8 gives the explicit coordinate identification, but does not use that citation to transfer singular-locus completeness. Nor does it transfer that completeness from Hassett–Tschinkel. Their Proposition 10 relates the determinantal hypersurfaces attached to a four-space in $\text{End}(V)$ and its trace orthogonal, a cubic threefold with six ordinary nodes in linear general position corresponding to a smooth paired cubic surface [16, §3, Proposition 10]; that statement is recorded as context and is not load-bearing. Singular-locus completeness is instead a kernel-checked theorem about the cross-golden family itself. Over any commutative ring containing a root of $t^2 = t + 1$,

```
RelativeConicArcs.SupportOrientationNodes.
derivative_crossGoldenDeterminantLine_eval
```

identifies the partial derivatives of the cross-golden determinant with the gradient quadrics of the support cubic, and over any field of characteristic zero containing such a root,

```
RelativeConicArcs.SupportOrientationNodes.  
singularPoints_crossGoldenDeterminant_eq_axisClasses
```

identifies the singular points with the six axis classes. Neither uses a smoothness or transversality hypothesis. The chartwise Gröbner bases displayed in the node-frame argument are the load-bearing finite human certificate; these declarations independently rederive the same result.

The line bound, chord-defect identity, partial-cover deduction, decoder reconstruction, support bipartition, and the orientation two-graph reconstruction are proved in the text. In particular, the conference identity gives pair balance and the Hessian ranks, while the five-matching normalizer and six-coset bound give the S_5/A_5 stabilizer distinction. The golden cross block, its four-dimensional trace complement, and the resulting gradient classification give singular-locus completeness without a smoothness input. The paper-owned replay independently constructs the coset actions, signed operator, triangle products, diagonal pencil, cross-golden trace dual, switching class, gradient ideals, and Hessian blocks. Its gradient exhaustion independently recomputes the displayed human certificate rather than replacing it. It reads no later-paper artifact:

```
check_orientation_two_graph.py.
```

None of the preceding manuscript conclusions is imported from the formal development. After the structural proofs are complete, the formal gate independently rederives the displayed q11 code and decoder conclusions, the causal containing-quadratic rigidity spine, the projective-to-code correspondence, the chord-defect specialization, the $q = 9$ obstruction, the small-arc moments, the intrinsic ten-plus-ten support split, the complete Brianchon matching dictionary, and the eight orientation mechanisms from the antipodal cover through the rational and integral commutants. Dye’s ten-point bound and his equality classification are theorems of the development rather than assumptions: the bound is proved for every field in which two is invertible and specialized to $q = 11$, and the classification is obtained from the golden normal form together with the two explicit projectivities carrying its order-eleven instances onto the displayed witness. The commutant theorems are unconditional: Lean checks golden equivariance on all sixty signed action matrices, proves the reverse containment from the displayed five-cycle and three-cycle commutation system, and checks the diagonal/off-diagonal integral test. The generated q11 rows are finite leaves internal to this redundant cross-check; they are not premises of the orbit–stabilizer or incidence proofs above. The companion searches lie outside this gate. The reusable Lean development is distributed in `finitegeom`, while the generated q11 tables are frozen in the `q11` package. Their proved compatibility with the human model is checked in the `Paper I bridge`. Neither upstream repository imports the other. The version-independent archival locator of `finitegeom` is the Zenodo concept DOI 10.5281/zenodo.21650878. The formalization uses Lean 4.32.0-rc1. The exact shared-library, certificate, and bridge commits are pinned once in `FORMAL_COMPANION.json`. Their verification boundaries are

```
human gate: RelativeConicArcs/Gates/ClebschRigidityTrust.lean  
certificate aggregate: TavisRuddFiniteGeom/Certificates/Q11.lean  
compatibility bridge:  
TavisRuddFiniteGeom/Papers/ClebschRigidity/CertificateCompatibility.lean  
Mathlib commit: 571b8a8e54219b4d393f75f4b8653fac08197fcc.
```

The SHA-256 digest of `verification/checker_outputs.json` is

98744378dee557bcd29a5ab5ab9f4fad827b62721c32558f47edefd06955397a.

The three repository manifests record the exact source and verification-boundary hashes checked by the release verifier. The two-generator word ledger used in Proposition 6.3 is a formal finite certificate; the remaining formalization is an independent cross-check rather than a manuscript proof dependency. Its recorded axiom audit lists the standard logical axioms alone for every audited terminal, and the gate uses neither admitted declarations nor native execution. In particular, the rational and integral commutant terminals have no proposition-valued representation-theoretic interface in their trust boundary.

10 Conclusion

For every fixed arc size, the partial-cover bound limits the field orders that can admit conic filling. Together with the universal defect identity and passant counts, it gives

$$2k - 3 \leq q \leq \frac{k(k-1) + 3}{3}.$$

Thus the conic-filling existence problem is finite for every fixed k . The Clebsch code realizes a different kind of sharpness: covering-radius data recover geometry that was not built into the code, forcing the Clebsch hexagon and hence its A_5 symmetry. A line bound and the defect identity reduce that rigidity to Dye’s equality case.

More generally, which odd prime powers in the quadratically widening window admit conic-filling arcs? A second problem is to connect fixed- k rigidity with the large-arc conic-forcing regime. Both ask how far a syndrome locus can reconstruct the projective system that produced it.

The same questions have useful algebraic and spectral forms. Passant conditions become clique problems in conic association schemes, while the decoder results ask when coset-leader support data alone reconstruct a projective code. For the Clebsch code, that support data also reconstructs the switching class of the golden continuation operator: the support cubic and the twelve-point syndrome locus retain the same orientation two-graph. The determinant pencil makes the bridge visible: its quadratic identity $B^2 = 5I$ controls every symmetric layer, while triangle holonomy is exactly the remaining cubic term. In particular, decoder support data recover a nontrivial integral endomorphism order, not only a projective configuration.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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